MNGOBI BHEKHE
BRIGHTLEARN TUTORIAL
EXERCISE 2

Database : employees_db

id	first_name	lastname	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

1. COUNT() FUNCTION

```
SELECT COUNT (*) AS total_employees
FROM employees;
```

total_employees
10

2. SUM() FUNCTION

SQL query to find the total salary paid to all employees in the IT department.

```
SELECT SUM (salary) As total_IT_salary
FROM employees
WHERE department = 'IT';
```

total_IT_salary
220 000

3. AVG() Function

SQL query to calculate the average salary of employees in the HR department

```
SELECT AVG(salary) AS avg_HR_salary
FROM employees
WHERE department = 'HR';
```

avg_HR_salary
49500

4. MIN() and MAX() Functions

SQL query to calculate the highest and lowest salary in the company.

```
SELECT MIN(salary) AS lowest_salary, MAX(salary) AS highest_salary
FROM employees;
```

lowest_salary	highest_salary
48000	62000

5 GROUPBY statement

SQL query to group employees by department and display the total salary paid in each department

```
SELECT department, SUM(salary) AS total_salary
FROM employees
GROUP BY department;
```

department	total_salary
Finance	119000
HR	99000
IT	220000
Marketing	105000

6. GROUP BY and COUNT()

SQL query to count how many employees work in each city.

```
SELECT city, COUNT(*) AS num_employees
```

FROM employees
GROUP BY city;

city	num_employees
Chicago	3
Houston	1
Los Angeles	2
New York	2
San Francisco	2

7. GROUP BY and ORDER BY

SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary

SELECT department, AVG (salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC;

department	avg_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

8. HAVING clause

SQL query to find departments where the total salary paid exceeds 100 000 (use GROUP BY and HAVING)

SELECT department, SUM (salary) AS total_salary
FROM employees

GROUP BY department
HAVING SUM (salary) > 100 000;

department	total_salary
Finance	119000
IT	220000
Marketing	105000

9. Combining GROUP BY, HAVING, and ORDER BY
SQL query to list cities where more than one employee works, ordered by the number of employees in descending

```
SELECT city, COUNT (*) AS num_employees
FROM employees
GROUP BY city
HAVING COUNT (*) > 1
ORDER BY num_employees DESC;
```

City	numemployees
chicago	3
Los Angeles	2
New York	2
San Francisco	2

10. Combining Aggregate Functions
SQL query to find the depart with the highest average salary.

```
SELECT department, AVG (salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC LIMIT 1;
```

department	avg_salary
Finance	59500